

Bash Scripting - Complete Guide

1. What is a Bash Script?

A Bash script is a text file containing Linux commands that can be executed by the Bash shell.

Example:

```
#!/bin/bash  
echo 'Hello, World!'
```

2. Types of Bash Scripts

1. Simple scripts - basic automation tasks.
2. Conditional scripts - use if/else.
3. Loop-based scripts - repeat tasks.
4. Functional scripts - modular structure.

3. Variables in Bash

Variables store data.

Syntax:

```
variable_name=value
```

Example:

```
name='Linux'  
echo 'Hello $name'
```

Types:

- Local Variables
- Environment Variables
- Positional Parameters (\$1, \$2, ...)
- Special Variables (\$?, \$\$, \$#)

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4. Operators in Bash

Operators perform operations on data.

Types:

- Arithmetic: +, -, *, /, %
- Relational: -eq, -ne, -gt, -lt, -ge, -le
- Logical: &&, ||, !
- String: =, !=, <, >, -z, -n
- File Test: -e, -f, -d, -r, -w, -x

5. Conditional Statements

Used to make decisions in scripts.

Syntax:

```
if [ condition ]; then
    commands
elif [ another ]; then
    commands
else
    commands
fi
```

Example:

```
if [ $age -ge 18 ]; then
    echo 'Adult'
else
    echo 'Minor'
fi
```

6. Loops in Bash

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Loops repeat commands.

Types:

1. For Loop
2. While Loop
3. Until Loop

Examples:

For Loop:

```
for i in {1..5}; do echo $i; done
```

While Loop:

```
count=1
```

```
while [ $count -le 5 ]; do echo $count; ((count++)); done
```

Until Loop:

```
count=1
```

```
until [ $count -gt 5 ]; do echo $count; ((count++)); done
```